# Chapter 7 - Superposition of Quasi-Parallel Plane Waves

- ↳ more complicated waves can be built → superposition
- ↳ phase velocity vs. group velocity
- ↳ Fourier theory
- ↳ dispersion

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$$\vec{E}(\vec{r},t) = \sum_j \vec{E}_j e^{i(\vec{k}_j \cdot \vec{r} - \omega_j t)}$$

amplitude + phase

$$\vec{B}(\vec{r},t) = \sum_j \vec{B}_j e^{i(\vec{k}_j \cdot \vec{r} - \omega_j t)}$$

amplitude + phase

$$= \sum_j \frac{\vec{k}_j \times \vec{E}_j}{\omega_j} e^{i(\vec{k}_j \cdot \vec{r} - \omega_j t)}$$

$$\vec{S}(\vec{r},t) = \mathrm{Re}\{\vec{E}(\vec{r},t)\} \cdot \frac{\mathrm{Re}\{\vec{B}(\vec{r},t)\}}{\mu_0}$$

time average to exclude many short variations

limit to k's that point mostly in the same direction

$$\langle \vec{S} \rangle = I = \frac{n\epsilon_0 c}{2} \cdot \vec{E}(\vec{r},t) \cdot \vec{E}^*(\vec{r},t)$$

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Group velocity vs. Phase Velocity

$$E_1 = \vec{E}_0 e^{i(\vec{k}_1 \cdot \vec{r} - \omega_1 t)} \qquad E_2 = \vec{E}_0 e^{i(\vec{k}_2 \cdot \vec{r} - \omega_2 t)}$$

$$v_{p1} = \frac{\omega_1}{k_1} \qquad v_{p2} = \frac{\omega_2}{k_2}$$

$$I(\vec{r},t) = \frac{n\epsilon_0 c}{2} \vec{E}_0 \cdot \vec{E}_0^* \left[ e^{i(\vec{k}_1 \cdot \vec{r} - \omega_1 t)} + e^{i(\vec{k}_2 \cdot \vec{r} - \omega_2 t)} \right] \left[ e^{-i(\vec{k}_1 \cdot \vec{r} - \omega_1 t)} + e^{-i(\vec{k}_2 \cdot \vec{r} - \omega_2 t)} \right]$$

$$I(\vec{r},t) = \frac{n\epsilon_0 c}{2} \vec{E}_0 \cdot \vec{E}_0^* \left( 2 + e^{i(\vec{k}_1 \cdot \vec{r} - \omega_1 t - \vec{k}_2 \cdot \vec{r} + \omega_2 t)} + e^{i(\vec{k}_2 \cdot \vec{r} - \omega_2 t - \vec{k}_1 \cdot \vec{r} + \omega_1 t)} \right)$$

$$I(\vec{r},t) = \frac{n\epsilon_0 c}{2} \vec{E}_0 \cdot \vec{E}_0^* \left( 2 + e^{i[(\vec{k}_2 - \vec{k}_1) \cdot \vec{r} - (\omega_2 - \omega_1) t]} + e^{-i[(\vec{k}_2 - \vec{k}_1) \cdot \vec{r} - (\omega_2 - \omega_1) t]} \right)$$

$$\cos(x) = \frac{e^{x} + e^{-x}}{2}$$

$$\sin(x) = \frac{e^{x} - e^{-x}}{2i}$$

$$I(\vec{r},t) = n\epsilon_0 c \cdot \vec{E}_0 \cdot \vec{E}_0^{*}\left(1 + \cos\left[\underbrace{(\vec{k}_2 - \vec{k}_1)}_{\Delta\vec{k}} \cdot \vec{r} - \underbrace{(\omega_2 - \omega_1)}_{\Delta\omega} t\right]\right)$$

$$v_p = \frac{\bar{\omega}}{\bar{k}}$$

$$\bar{\omega} = \frac{\omega_1 + \omega_2}{2}$$

average

$$\bar{k} = \frac{k_1 + k_2}{2}$$

$$v_g = \frac{\Delta\omega}{\Delta k} = \frac{d\omega}{dk}$$